WORKFORCE360

EMPLOYEE LEAVE & ATTENDANCE MANAGEMENT SYSTEM PROJECT REPORT

Built using Microsoft Power Platform

Microsoft Dataverse • Power Apps • Power Automate • Microsoft Outlook

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Project-specific implementation note: This report reflects the actual Workforce360 setup described during development: the Dataverse HR tables, the Workforce360 Employee Portal canvas app, the Workforce360 HR Management model-driven app, the leave approval flow, and the Leave Request date-validation business rule. Any functionality not confirmed as tested is identified as pending verification.

1. EXECUTIVE SUMMARY

Workforce360 is an employee leave and attendance management solution being developed in Microsoft Power Platform. Its purpose is to organise employee information, leave applications, leave types, leave balances, attendance records, company holidays, attendance exceptions, approval history, and policy settings in one Dataverse environment.

The project uses Dataverse tables as the central data store. The canvas app, named “Workforce360 Employee Portal,” provides an employee-oriented entry point and includes a home screen named scrHome with the heading “Welcome to Workforce360” and navigation buttons such as Apply for Leave, My Leave Requests, My Attendance, and My Profile. A separate model-driven app, “Workforce360 HR Management,” has been created and the project tables have been added to its navigation for administrative access.

A Power Automate flow named “Workforce360 - Leave Approval” has been configured around a Dataverse-triggered leave approval process. It uses an approval action and a condition that checks the approval Outcome against “Approve.” The approval path updates the request to Approved, while the other path updates it to Rejected; email notification actions are part of the intended communication process and should be tested against the correct employee email field.

A Dataverse business rule on Leave Request has also been activated to prevent an End Date earlier than the Start Date by showing an error on the End Date field. The solution is an iterative project rather than a confirmed production deployment: the final data-source mappings, approval outcomes, email delivery, form behaviour, security, and publishing should be verified in the target environment.

2. PROJECT BACKGROUND, PROBLEM AND OBJECTIVES

2.1 Background

Leave and attendance administration requires consistent employee records, leave request details, decision tracking, and attendance information. When these activities are spread across email threads and spreadsheets, it becomes harder to identify the current status of a request, keep records consistent, and provide HR with a consolidated view.

2.2 Problem addressed

Workforce360 addresses the need for a single structured place to maintain employee and HR records, capture leave requests, route decisions, record outcomes, and support HR administration. It is designed to reduce manual follow-up and make the request lifecycle easier to track.

2.3 Project objectives

• Maintain employee records and associated HR data in Dataverse.

• Capture leave requests with leave type, start date, end date, reason, and status.

• Route submitted leave requests through an approval workflow.

• Update the request status based on the approver's decision.

• Support email communication for approval or rejection.

• Provide employee-oriented navigation through the canvas app.

• Provide HR-oriented table management through the model-driven app.

• Apply validation so the End Date cannot be earlier than the Start Date.

2.4 Intended users

Employees use the portal to access leave-related functions; managers/approvers make leave decisions; HR staff maintain records and review requests; and environment administrators maintain application configuration and access.

3. PROJECT SCOPE AND REQUIREMENTS

3.1 In-scope modules

The Dataverse data model was planned around the following tables: Employee, Leave Type, Leave Request, Attendance, Leave Balance, Company Holiday, Attendance Exception, Approval History, and Policy Settings. The model-driven app navigation has been populated with the project tables.

3.2 Employee portal

The canvas app is named “Workforce360 Employee Portal.” Its home screen is scrHome and displays “Welcome to Workforce360.” The employee-facing navigation includes Apply for Leave, My Leave Requests, My Attendance, and My Profile. Check-In and Check-Out buttons were explored during development but removed while the related formulas and flow connection were unresolved; they should not be described as completed features.

3.3 HR management

The model-driven app is named “Workforce360 HR Management.” Its purpose is to provide a table-driven interface for HR administration. The project tables were added to its navigation. Views, dashboards, security roles, and final publication should be checked before they are represented as complete deliverables.

3.4 Functional requirements

FR1: Store employee and HR records in Dataverse.

FR2: Capture leave request information and its status.

FR3: Trigger an approval workflow for an applicable leave request.

FR4: Update the leave request according to the approval outcome.

FR5: Support approval/rejection email notifications after correct recipient mapping and testing.

FR6: Validate the date range on Leave Request.

FR7: Allow HR users to access the relevant tables through the model-driven app.

3.5 Out of scope or pending verification

Automatic attendance check-in/check-out, automatic leave balance deduction/accrual, holiday-aware working-day calculation, comprehensive role-based security, and production deployment are not confirmed as completed. They are potential future enhancements unless subsequently implemented and tested.

4. TECHNOLOGY AND SOLUTION ARCHITECTURE

4.1 Microsoft Dataverse

Dataverse is the structured data layer. It stores records in related tables and provides forms, views, relationships, and business rules. The exact logical/schema names should be confirmed in the target environment because duplicate tables were accidentally created during setup. The connected data source shown in Power Apps and the table selected in each flow action must be checked to ensure the intended table is used.

4.2 Power Apps canvas app

The canvas app, Workforce360 Employee Portal, is intended for employee self-service. It contains the scrHome screen and employee navigation. Canvas formulas depend on the exact connected data source and Dataverse column names; therefore, formula-based features should be marked complete only after a successful end-to-end test.

4.3 Power Apps model-driven app

Workforce360 HR Management is the HR-facing app. It was created in the same solution/environment and the project tables were added to navigation. Model-driven apps use Dataverse forms, views, and relationships to support record management.

4.4 Power Automate

The leave approval flow starts from a Dataverse trigger, presents an approval action, checks the approval Outcome, and updates the request status. The approval outcome comparison must use the actual returned Outcome value, with the approved and rejected branches mapped to the intended status values.

4.5 Outlook email

Outlook email actions can communicate the decision to the employee. The recipient should be taken from the correct employee work-email field or the email field captured on the request. Test both approval and rejection paths, and verify the sent message and recipient before claiming email delivery is fully operational.

4.6 Architecture flow

Employee / HR user → Power Apps interface → Dataverse record → Power Automate approval → Dataverse status update → Outlook notification (after configuration and testing).

5. DATAVERSE DATA MODEL

The project's planned data model contains nine tables. The final lookup and relationship configuration should be verified in Dataverse before deployment.

Employee: employee profile information such as name, work email, department, job title, manager, joining date, and employment status.

Leave Type: leave categories and associated descriptions or policy attributes.

Leave Request: the main transaction table for employee leave applications. It is expected to reference the employee and leave type and hold start date, end date, requested days, reason/comments, status, and decision information.

Attendance: daily attendance records and related time/status information.

Leave Balance: leave entitlement or available balance information associated with an employee and leave category.

Company Holiday: dates and descriptions of company holidays.

Attendance Exception: records requiring HR review, such as missing or irregular attendance information.

Approval History: a trace of approval decisions, approver details, timestamps, and comments where configured.

Policy Settings: organisation-level settings that can support leave or attendance policy rules.

Important implementation detail: duplicate tables were created accidentally during development. For example, tables with suffixes such as "Employee11," "Employee45," "Attendance1," and "Leave Request1" may exist alongside the intended tables. Do not delete records or tables without checking dependencies. Confirm the intended table in the app's Data pane and in each Power Automate trigger/action, then use one consistent table throughout the solution.

6. IMPLEMENTED VALIDATION AND BUSINESS RULE

6.1 Leave date validation

A business rule was created on the Leave Request table and activated. Its intended logic is:

IF End Date is earlier than Start Date

Show an error message on the End Date field:

“End Date cannot be earlier than Start Date.”

ELSE

Take no action.

This rule protects the request from an invalid date range when the rule is applied to the relevant Dataverse form and the form is published. Activation was reported during development; final confirmation should include opening the relevant form and attempting to enter an End Date earlier than the Start Date.

6.2 Why this rule matters

A reversed date range can create confusing requests and inaccurate duration calculations. A field-level error gives the user immediate feedback before the record is treated as valid.

6.3 Additional validation opportunities

The following are recommendations, not confirmed completed rules: require rejection comments when a request is rejected; make key employee and leave request fields mandatory; prevent duplicate attendance entries for the same employee and date; and validate that requested dates comply with organisational policy.

6.4 Business rule limitations

Business rules must be tested in the relevant form and app context. A business rule alone does not prove that every integration, imported record, or custom client formula is protected. Additional validation may be needed for data written through flows or APIs.

7. LEAVE APPROVAL AUTOMATION

7.1 Purpose

The “Workforce360 - Leave Approval” flow is intended to reduce manual follow-up by routing a leave request to an approver and updating the request after a decision.

7.2 Configured process

1. A Dataverse trigger starts the flow when the configured leave-request event occurs.

2. The flow runs an approval action.

3. A Condition checks whether the approval Outcome equals “Approve.”

4. In the True branch, the request status is updated to Approved.

5. In the False branch, the request status is updated to Rejected.

6. Email actions are used/intended to inform the employee of the decision, subject to confirming their configuration and recipient mapping.

7.3 Critical configuration checks

• Confirm the trigger monitors the intended Leave Request table, not an accidental duplicate.

• Confirm the trigger conditions prevent unwanted repeat runs if the flow updates the same row.

• Confirm the Condition compares the approval action’s Outcome value to the exact expected value “Approve.”

• Confirm the Update a row actions use the correct Dataverse table and the row ID from the trigger.

• Confirm the status column accepts the chosen Approved and Rejected values.

• Confirm email To is mapped to the correct employee email.

• Test one approved request and one rejected request, then inspect the Dataverse row and the email received.

7.4 Expected result

The request status should reflect the approver’s decision and the employee should receive the matching notification when the email actions are correctly configured. The flow was reported as running correctly during development; this report does not assume that every branch and email has been independently verified in production.

8. APPLICATION INTERFACES AND USER JOURNEYS

8.1 Workforce360 Employee Portal

The canvas app is the employee-facing entry point. The current home screen is scrHome and displays “Welcome to Workforce360.” The navigation buttons created include Apply for Leave, My Leave Requests, My Attendance, and My Profile. These buttons provide the intended structure for employee self-service; each destination screen and its data operations should be tested to ensure it performs the expected action.

8.2 Workforce360 HR Management

The model-driven app provides HR-oriented access to Dataverse records. All project tables were added to its navigation according to the development update. The next verification step is to open each page, confirm the correct table is shown, and test creating, opening, editing, and saving a record.

8.3 Suggested HR views

Useful views include Pending Leave Requests, Approved Leave, Rejected Leave, Active Employees, Attendance Today, Open Attendance Exceptions, and Upcoming Company Holidays. These should be labelled as configured only after they have been created and tested.

8.4 User journeys

Employee: open the portal → start a leave request → enter the leave details → submit → receive the decision notification after approval.

Approver: receive the approval request → choose Approve or Reject → flow updates the status.

HR: open Workforce360 HR Management → select the relevant table → review or maintain the record.

8.5 Check-in/check-out status

Check-In and Check-Out buttons were attempted but removed because the formulas and flow connection were not resolved. They are not included as completed functionality in this report. A future version can add an attendance action that creates or updates the correct Attendance record and optionally sends a confirmation email.

9. TESTING, SECURITY AND DEPLOYMENT

9.1 Recommended acceptance tests

Test 1 — Valid leave dates: enter Start Date earlier than or equal to End Date and verify the form accepts the range.

Test 2 — Invalid leave dates: set End Date earlier than Start Date and verify the business rule shows the configured error.

Test 3 — Approval: create a test request, approve it, and verify the same Dataverse row becomes Approved.

Test 4 — Rejection: create a separate test request, reject it, and verify the row becomes Rejected.

Test 5 — Email: confirm that each test decision sends the correct message to the intended employee.

Test 6 — Correct table mapping: confirm app screens and flows use the intended tables rather than duplicate tables.

Test 7 — Model-driven app: open each navigation item and create or edit a test record where appropriate.

Test 8 — Permissions: test access with the intended employee, manager, HR, and administrator roles after roles are configured.

9.2 Security

Use least-privilege access. Employees should only access records appropriate to them; managers should access requests they are authorised to review; HR should have the access needed for administration; and administrators should manage configuration. These are recommended access boundaries and must be implemented using security roles, table privileges, and, where necessary, row-level access controls.

9.3 Deployment checklist

• Verify table names, relationships, columns, and choice values.

• Resolve duplicate-table references carefully.

• Publish the business rule and relevant forms.

• Confirm the flow is enabled and its connection references are valid.

• Test both approval outcomes and email delivery.

• Configure security roles and test with non-admin accounts.

• Publish the canvas and model-driven apps.

• Keep test data separate from real employee records.

9.4 Known project risks

Duplicate tables can cause flows or apps to read and update the wrong data. Incorrect field mapping can prevent emails from being delivered. Unpublished rules may not behave as expected. Overly broad permissions can expose employee information. These risks should be addressed before real employee data is used.

10. CONCLUSION AND FUTURE ENHANCEMENTS

10.1 Conclusion

Workforce360 establishes a Power Platform foundation for employee leave and attendance management. The project has a Dataverse-based HR data model, a canvas app named Workforce360 Employee Portal, a model-driven app named Workforce360 HR Management, a configured leave approval flow, and an activated Leave Request date-validation business rule. These components address the core need for centralised records, structured leave requests, decision tracking, and HR administration.

The next priority is not to add more features immediately, but to verify the existing solution end to end: ensure the apps and flow use the intended tables, test both approval outcomes, confirm email delivery, validate the date rule on the published form, and configure security.

10.2 Future enhancements

• Complete employee check-in/check-out with reliable attendance record creation.

• Calculate leave duration using working days and company holidays.

• Update leave balances after approval and define cancellation/reversal behaviour.

• Create HR dashboards and operational views.

• Require rejection comments and record a complete approval history.

• Add role-based security and row-level access.

• Improve employee notifications and error handling.

• Add audit, monitoring, and a documented deployment process.

10.3 Final project status

The solution is a working development build with key components configured. It should be described as production-ready only after the acceptance tests, permissions, email delivery, and publication checks have passed.

References

1. Microsoft Learn — Power Apps documentation: https://learn.microsoft.com/power-apps/

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3. Microsoft Learn — Power Automate documentation: https://learn.microsoft.com/power-automate/

4. Microsoft Learn — Business rules in model-driven apps: https://learn.microsoft.com/power-apps/maker/data-platform/data-platform-create-business-rule